



Operating Systems
CS 315 - 040
Department of Physics and Computer Science
Medgar Evers College
Exam 3

Instructions:

- The exam requires completing a set of tasks within 120 minutes.
- For each task, write your solution in a separate file. Tasks 1 - 9 must be text files. Each file must be named 'Task0 n ' where n is its task number.
- For problems 5 through 9, you must complete the specified scheduler timeline. Then, provide a table that includes the completion time, turnaround time, and waiting time for each process, where

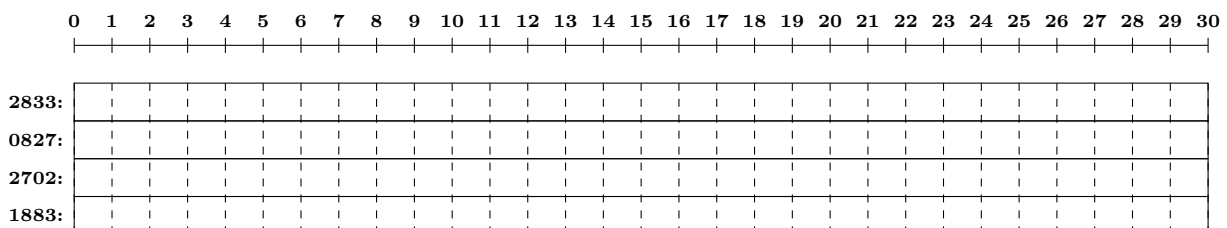
$$\begin{aligned}\text{turnaround time} &= \text{completion time} - \text{arrival time} \\ \text{waiting time} &= \text{turnaround time} - \text{burst time}\end{aligned}$$

- Use the IAS instruction set when necessary:
- Notes are not allowed.
- Cheating of any kind is prohibited and will not be tolerated.
- Violating and/or failing to follow any of the rules will result in an automatic zero (0) for the exam.

TO ACKNOWLEDGE THAT YOU HAVE READ AND UNDERSTOOD THE INSTRUCTIONS ABOVE,
PRINT YOUR NAME AND THE DATE ON YOUR SUBMISSIONS

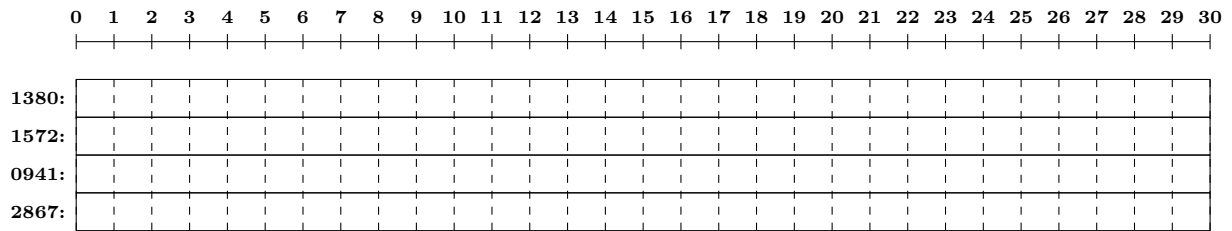
Grading

Section	Maximum Points	Points Earned
1	2.0	
2	3.0	
3	3.0	
4	2.0	
5	2.0	
6	2.0	
7	2.0	
8	2.0	
9	2.0	
Total	20	



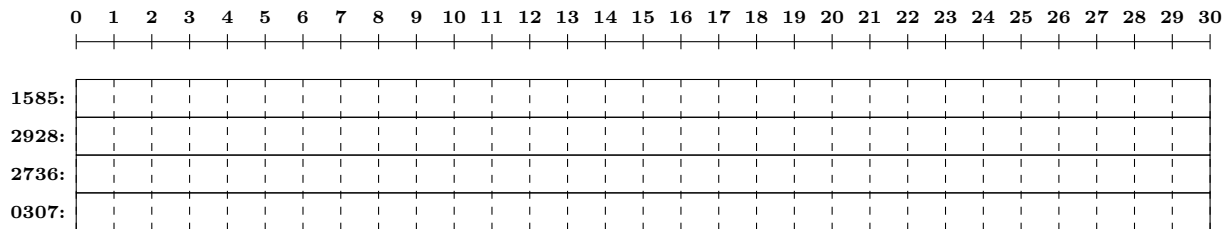
7. Using the given processes, simulate the Shortest Job First (SJF) Scheduler

PID:	1380	1572	0941	2867
Arrival Time:	0	2	6	4
Burst Time:	5	6	8	3



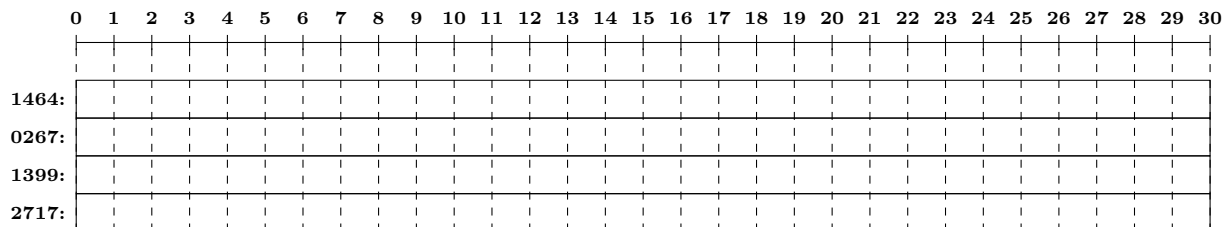
8. Using the given processes, simulate the Round-Robin (RR) Scheduler using time quantum = 3

PID:	1585	2928	2736	0307
Arrival Time:	0	3	6	9
Burst Time:	7	5	8	4



9. Using the given processes, simulate the Shortest Remaining Time First (SRTF) Scheduler

PID:	1464	0267	1399	2717
Arrival Time:	2	4	6	0
Burst Time:	3	6	4	5



IAS Instruction Set Reference

Mnemonic	Opcode	Description
LMA	0A	Transfer contents from MQ to AC
LDM	09	Transfer M(X) to MQ
STA	21	Transfer contents from AC to memory location X
LDA	01	Transfer M(X) to AC
LDN	02	Transfer -M(X) to AC
ALD	03	Transfer M(X) to AC
ALN	04	Transfer - M(X) to AC
BRL	0D	Takes next instruction from left half of M(X)
BRR	0E	Takes next instruction from right half of M(X)
BPL	0F	If AC ≥ 0 , takes next instruction from the left half of M(X)
BPR	10	If AC ≥ 0 , takes next instruction from the right half of M(X)
ADD	05	Add M(X) to AC; put result in AC
AAD	07	Add M(X) to AC; put result in AC
SUB	06	Subtract M(X) from AC; put result in AC
ASB	08	Subtract M(X) from AC; put result in AC
MUL	0B	Multiply M(X) by MQ; put most significant bits of result in AC; least significant in MQ
DIV	0C	Divide AC by M(X); put quotient in MQ and remainder in AC
LSH	14	Multiply AC by 2
RSH	15	Divide AC by 2
STL	12	Transfer AC[28:39] to M(X)[8:19]
STR	13	Transfer AC[28:39] to M(X)[28:39]
HLT	00	Halts